of a player's cards). This joint recommendation is then encoded by summing the indices for the individual recommendations and using modular arithmetic. The encoded joint recommendation is mapped to different hints that HatBot could make, specifically, whether it reveals the color or rank of a card for each other player. Since each player can view everyone's cards but their own, they can reconstruct the action recommended to them by figuring out what would have been recommended to the other players based on HatBot's convention, which HatBot assumes they know and use. Although this convention is not very intuitive for human players, it would still be possible for humans to learn and follow. Cox and colleagues also introduce an "information strategy" using a similar encoding mechanism to directly convey information about each player's cards (as opposed to a recommended action), however it requires additional bookkeeping that makes it impractical for humans to use. As originally proposed, both the recommendation and information strategies were tailored for playing 5-player Hanabi. However, a variant of the information strategy, called *WTFWThat* [41], can play two through five players. These benchmarks more provide a lower-bound for optimal play than a baseline suggestive of human performance.

FireFlower [42]. FireFlower implements a set of human-style conventions (detailed in Appendix A.1). The bot keeps track of both private and common knowledge, including properties of cards that are implied by the common knowledge of what the conventions entail. Using this, FireFlower performs a 2-ply search over all possible actions with a modelled probability distribution over what its partner will do in response, and chooses the action that maximizes the expected value of an evaluation function. The evaluation function takes into account the physical state of the game as well as the belief state. For example, if there is a card in the partner's hand that is commonly known (due to inference from earlier actions) to be likely to be playable, then the evaluation function's output will be much higher if it is indeed playable than if it is not. FireFlower uses a few additional conventions for three and four players, but avoids the hat-like strategies in favour of conventions that potentially allow it to partner more naturally with humans. According to Fireflower's creator, it is designed with a focus on maximising the win probability, rather than average score.

4.3. Experimental Results: Self-Play

Table 1 shows the experimental results of the baseline agents and state-of-the-art learning algorithms with each number of players. First, note that neither the Rainbow nor ACHA agent reaches the performance of the best hand-coded agent in the two-player setting (SmartBot), and neither learning agent reaches the performance of any of the hand-coded agents with more than two players. There is a large performance gap in what is possible (as demonstrated by the hand-coded agents) and what state-of-the-art learning algorithms achieve. Even rule-based strategies that codify more human-like conventions achieve scores higher than the learning algorithms, particularly in the three and five player setting. Experienced human teams are generally considered stronger than such bots, suggesting the gap between these learning agents and superhuman performance in self-play is even larger still.

The final learning agent, BAD, only reports results for the two player setting, but achieves the best reported performance in that setting. Because BAD was only tested on two player Hanabi and a small synthetic game, we can not make a strong claim that BAD is an algorithmic improvement for a general class of games, rather than just being

Regime	Agent	2P	3P	4P	5P
–	SmartBot	22.99 (0.00) 29.6%	23.12 (0.00) 13.8%	22.19 (0.00) 2.076%	20.25 (0.00) 0.0043%
–	FireFlower	22.56 (0.00) 52.6%	21.05 (0.01) 40.2%	21.78 (0.01) 26.5%	-
–	HatBot	–	–	–	22.56 (0.06) 14.7 %
–	WTFWThat	19.45 (0.03) 0.28%	**24.20** (0.01) **49.1**%	**24.83** (0.01) **87.2**%	**24.89** (0.00) **91.5**%
SL	Rainbow	20.64 (0.11) 2.5 %	18.71 (0.10) 0.2%	18.00 (0.09) 0 %	15.26 (0.09) 0 %
UL	ACHA	22.73 (0.12) 15.1%	20.24 (0.15) 1.1%	21.57 (0.12) 2.4%	16.80 (0.13) 0%
UL	BAD	**23.92** (0.01) **58.6**%	–	–	–

Table 1: Shown are the results for the three learning agents, Rainbow, ACHA and BAD, compared to the rule-based agents, SmartBot, FireFlower and HatBot, for different numbers of players in *self-play*. For each algorithm and number of players we show mean performance of the best agent followed by (standard error of the mean) and percentage of perfect (*i.e.*, 25 point) games. Error of the mean differs based on different number of evaluation games.

good at Hanabi. However, we find it suggestive that BAD's tracking of all agents' beliefs leads to a marked improvement in performance for an agent learning to play.

Notice that both BAD and the ACHA agent require vast amounts of training data to establish good performance. As such, the self-play challenge, even for two player Hanabi, is fertile ground for further innovation. Furthermore, theory-of-mind-inspired methods still need to be scaled to settings with more than two players.

Comparing the more traditional RL approaches, the ACHA agent in the unlimited regime (using over 20 billion steps of experience for each learner in the population) achieved higher scores than Rainbow (using 100 million steps of experience) across all number of players. This quite naturally may be due to ACHA using more training experience, but may also be due to Rainbow's feed-forward network architecture with no history of past actions possibly making it harder to learn multi-step conventions. Both agents saw a decline in performance as the number of agents increased with Rainbow's more gradual, while ACHA saw a precipitous drop in performance with five players.

Figure 2 shows training curves for one run of ACHA showing the performance of the multiple "agents" within the population. Note that these curves are linked together through parameter evolution and are not independent. In all but the four player setting, it appears ACHA has found a local minimum in policy space and it is unable to escape even with more training. However, parameter evolution is hiding the true extent of the local minima problem. Figure 3 gives ACHA training curves for two and four player games when evolution is disabled, so each curve is an independent self-play trial, using the same fixed hyper-parameters found as the best in the earlier experiment. Here we can see that independent learning trials find a wide array of different local minima, most of which are difficult to escape. For example, in the two player setting, roughly a third of the independent agents are below 15 points and appear to no longer be improving.

We further find that even ACHA runs with similar final performance can learn very different conventions. For example, one agent uses color hints to indicate that the 4th card of the teammate can likely be discarded, while another agent uses the different color

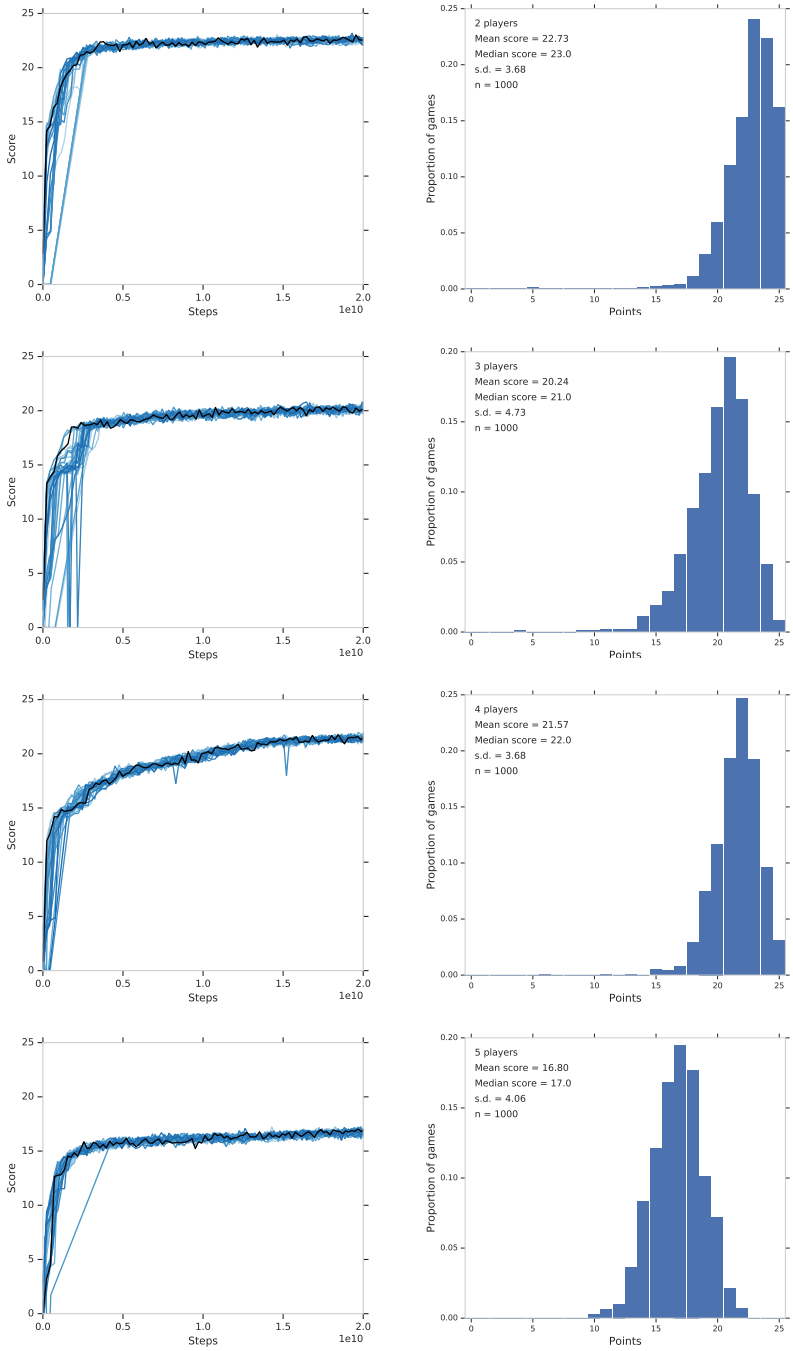

Figure 2: ACHA results for two to five players, from top to bottom respectively. Performance curves (left) are training-time results from the current soft-max policy. Average scores and distributions (right) are test-time results from 1000 episodes generated using the greedy policy from the agent with the best training score.